

# QUESTION-TO-SLIDE MAPPING

## SECTION A - LECTURE 02

### Climate & Built Environment

|               |                                                                                                                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Course        | BECM 3115 - Climate and Architectural Design                                                                                                                                                                            |
| Lecture deck  | 2.0 - Climate & Build Environment (15 PDF pages/slides)                                                                                                                                                                 |
| Question bank | Regular term-final examinations, 2016-2024 (8 supplied papers; 16 PDF pages)                                                                                                                                            |
| Method        | Question wording and answer-content matched to specific lecture slides. A link is counted only when the deck supplies usable answer content; a slide that merely raises a question is not treated as a complete answer. |

#### RESULT

13 past-question links were found across four examination years, representing 165 mapped marks. The strongest directly covered family is KHANA's climate-responsive zoning (2017, 2018 and 2024). Climate-vs-Environment and Vitruvius are also directly covered; designer-responsibility and contemporary/traditional-comparison questions are only partly supported and must be completed from later lectures.

Prepared as part of a cumulative lecture-by-lecture question bank analysis.

# 1. Executive Summary

Lecture 02 links climate-responsive design to historical and vernacular precedents. Its best exam return comes from the KHANA block, the Climate-vs-Environment definition, and Vitruvius. Historical exam section labels vary, so the mapping below follows subject matter and the current Section A classification rather than old printed A/B labels.

|                 |                |               |                |                 |              |
|-----------------|----------------|---------------|----------------|-----------------|--------------|
| 15              | 13             | 3             | 2              | 8               | 165          |
| Slides reviewed | Question links | Exact matches | Direct matches | Partial matches | Mapped marks |

| Classification | Meaning                                                                                                                |
|----------------|------------------------------------------------------------------------------------------------------------------------|
| Exact          | The requested definition/list/diagram appears essentially as asked and can be answered directly from the cited slides. |
| Direct         | Most or all answer content is present; the student must justify, synthesize or apply the slide material.               |
| Partial        | Only one component or supporting framework is present; another lecture/source is required for a complete answer.       |

Section-label note

The source deck cover says “Climate & Design (B)”. It is treated here as Section A - Lecture 02 because that is how the lecture was supplied and organized for the current course preparation. Historical exam questions are mapped by content, not by their old printed section label.

Slide-numbering note

The mapping uses PDF page/slide numbers with the title page as Slide 1. In this deck, the visible slide numbers from Slide 2 onward match the PDF page count.

## Highest-priority slide blocks

| Priority | Slides | Topic                                                     | Reason                                                                                                                                                       |
|----------|--------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 9-12   | KHANA maxim, climatic zoning and contemporary application | Three-paper cluster (2017, 2018, 2024). Slides 9-11 provide the actual zoning logic and sketch; Slide 12 signals contemporary application.                   |
| 2        | 6-7    | Vitruvius: climate and architecture                       | 2017 asked the suggestions directly. Slide 7 is almost an exam-ready eight-point list.                                                                       |
| 3        | 2      | Climate vs Environment                                    | Exact 2017 short question and partial support for broader Climate-Context-Comfort comparisons.                                                               |
| 4        | 13-14  | Then-vs-now climate response + designer task              | Repeated responsibility/complexity and traditional-vs-contemporary questions draw on this comparison, but the slides do not provide complete formal answers. |
| 5        | 4-5    | Aristotle + Xenophon                                      | Strong conceptual evidence that climate-responsive design is not new; no direct 2016-2024 question found from these names.                                   |
| 6        | 8      | Vastu Shastra                                             | Shown as a climate-responsive planning graphic; no explanatory text and no direct past-question match found.                                                 |

## 2. Detailed Year-wise Question Mapping

Question wording has been lightly standardized for readability without changing the assessed meaning. "QB p." is the PDF page in the uploaded 2016-2024 regular question bank.

### Examinations 2016-2018

| Year | Original    | Marks | QB p. | Question / task                                                                                                                                     | Match   | Lecture slides | Coverage assessment                                                                                                                                                                                                               |
|------|-------------|-------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2017 | Sec B Q5(a) | 10    | 2     | Explain "Climate" and "Context" from the point of view of a built-form designer.                                                                    | Partial | 2              | Slide 2 defines Climate and Environment but does not define Context. Use Lecture 01 for the Context half.                                                                                                                         |
| 2017 | Sec B Q5(b) | 15    | 2     | According to KHANA, the south-facing house is best, east second, north a burden and west a curse. Do you agree? Justify with examples and sketches. | Direct  | 9-12           | Slides 9-11 give the exact maxim, zoning sketch and climatic reasoning for east pond, west bamboo, north banana/low planting and an open south. Slide 12 prompts contemporary application.                                        |
| 2017 | Sec B Q5(c) | 10    | 2     | Write down the suggestions of Vitruvius on climate and architecture.                                                                                | Exact   | 6-7            | Slide 6 introduces Vitruvius; Slide 7 lists the eight climatic suggestions almost exactly in answer form.                                                                                                                         |
| 2017 | Sec B Q6(a) | 5     | 2     | What is the difference between Climate and Environment?                                                                                             | Exact   | 2              | Slide 2 directly gives both definitions and distinguishes overall physical environment from the climate portion perceived through thermal sensation.                                                                              |
| 2017 | Sec B Q6(c) | 20    | 2     | What are the responsibilities of the built-form designers? What kind of complexities do they have to face in Climate Design?                        | Partial | 13-14          | Slide 13 compares past and present issues (cost, health, power/system failure, internal climate); Slide 14 asks the designer's task but does not list formal responsibilities or complexities. Lecture 04 is the complete source. |
| 2018 | Sec A Q1(a) | 10    | 3     | What differences exist among Climate, Context and Comfort?                                                                                          | Partial | 2              | Only Climate and Environment are defined here. Context and Comfort are not developed in Lecture 02; use Lecture 01 for the full three-way comparison.                                                                             |
| 2018 | Sec A Q1(b) | 15    | 3     | Defend your position: should Climate and Built Environment accept fate, reject development, or mitigate?                                            | Partial | 13-14          | The "then vs now" table and "designer's task" prompt help frame why blind dependence on artificial microclimate has costs and failure risks, but the exact accept/reject/mitigate argument is in Lecture 01.                      |
| 2018 | Sec A Q2(a) | 15    | 3     | Show the zoning of a building on a particular site according to KHANA for a climate-responsive building in Bangladesh, with necessary sketches.     | Exact   | 9-11           | Slides 9-11 provide the zoning arrangement, sketch and environmental interpretation for all four directions.                                                                                                                      |

#### 2016 diagnostic

No 2016 question was counted as a meaningful Lecture-02 answer-content match. The 2016 short note on Context, Comfort and Man-made Environment belongs to Lecture 01; Slide 2 here only gives the broader Environment/Climate distinction.

## 2. Detailed Year-wise Question Mapping (continued)

### Examinations 2020-2023

| Year | Original    | Marks | QB p. | Question / task                                                                                                                    | Match   | Lecture slides | Coverage assessment                                                                                                                                                                                            |
|------|-------------|-------|-------|------------------------------------------------------------------------------------------------------------------------------------|---------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020 | Sec B Q5(a) | 15    | 5     | What are the responsibilities of the built-form designers? What kinds of complexities do they have to deal with in climate design? | Partial | 13-14          | The table on Slide 13 identifies cost, health, power and system-failure concerns; Slide 14 asks the designer's task. It does not state the formal responsibility/complexity lists; use Lecture 04.             |
| 2020 | Sec B Q5(b) | 15    | 5     | Evaluate and compare contemporary and traditional building materials from the climatic point of view.                              | Partial | 13             | Slide 13 compares "then" and "now" on material transport, internal climate, cost, health and failure modes. It is relevant context, but it does not provide a proper material-by-material climatic comparison. |

|                 |                                                                                                                                                                                                                             |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021 diagnostic | No 2021 question was counted as a meaningful Lecture-02 answer-content match. The paper focuses on global climate, tropical climate, site/urban climate, thermal comfort and ventilation.                                   |
| 2022 diagnostic | No 2022 question was counted as a meaningful Lecture-02 answer-content match. Climate-responsive themes recur, but the usable answer content is in the tropical-climate, microclimate, comfort and ventilation lectures.    |
| 2023 diagnostic | No 2023 question was counted as a meaningful Lecture-02 answer-content match. The recent sustainability case has some thematic overlap with Slide 13, but this deck does not give enough direct answer content to count it. |

## 2. Detailed Year-wise Question Mapping (continued)

### Examination 2024

| Year | Original    | Marks | QB p. | Question / task                                                                                                                                                                                                 | Match   | Lecture slides | Coverage assessment                                                                                                                                                                                                                                               |
|------|-------------|-------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | Sec B Q5(a) | 10    | 13    | Explain the differences among climate, context and comfort in architectural design. Discuss how each factor influences design decisions.                                                                        | Partial | 2              | Slide 2 gives the Climate/Environment distinction only. Context, Comfort and their design-decision framework are not taught here; use Lecture 01 for the complete answer.                                                                                         |
| 2024 | Sec B Q5(b) | 15    | 14    | Discuss the relevance of KHANA's maxim in contemporary architectural practice. How can its environmental and climatic insights be applied to functional zoning and layout planning of a contemporary residence? | Direct  | 9-12           | Slides 9-11 give the environmental logic and zoned plan; Slide 12 explicitly asks about application in a contemporary residence. The student must translate the vernacular logic into a modern layout - the slide does not supply a ready-made contemporary plan. |
| 2024 | Sec B Q6(b) | 10    | 14    | Discuss the designer's responsibilities in ensuring thermal comfort in built environment and the domains of essential knowledge a designer should have.                                                         | Partial | 13-14          | Slide 13 gives consequences of artificial climate control and Slide 14 raises the designer-task question, but neither provides the required responsibility or essential-knowledge domains. Use Lecture 04 and thermal-comfort lectures.                           |

**Current-paper  
signal**

The 2024 paper returned directly to the KHANA block with a 15-mark application question. This is the clearest recent signal for Lecture 02: master Slides 9-11 as the evidence base, then be ready to apply that logic to contemporary functional zoning.

### 3. Repeated Question Patterns

A repeated cluster means the tested definition, decision or application is substantially the same. The “coverage / exam insight” column shows whether Lecture 02 can actually answer it.

| # | Pattern                                                         | Years / questions                         | Slides | Coverage / exam insight                                                                                                                     |
|---|-----------------------------------------------------------------|-------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | KHANA maxim / zoning / contemporary application                 | 2017 Q5(b); 2018 Q2(a); 2024 Q5(b)        | 9-12   | Three-paper cluster and the highest-return block. Learn both the four-direction layout and the environmental reasoning.                     |
| 2 | Designer responsibilities / climate-design complexities         | 2017 Q6(c); 2020 Q5(a); 2024 Q6(b)        | 13-14  | Three appearances, but only partial coverage. Slide 13 gives the problem context; Lecture 04 gives the formal answer lists.                 |
| 3 | Climate vs Environment / broader CCC comparison                 | 2017 Q5(a), Q6(a); 2018 Q1(a); 2024 Q5(a) | 2      | Slide 2 completely answers Climate-vs-Environment, but it does not define Context or Comfort.                                               |
| 4 | Vitruvius and climatic design                                   | 2017 Q5(c)                                | 6-7    | Single direct appearance, but very easy marks because Slide 7 is already an eight-point answer.                                             |
| 5 | Traditional vs contemporary climate response / mitigation logic | 2018 Q1(b); 2020 Q5(b)                    | 13-14  | Useful support only. The table compares “then” and “now” but does not provide the full mitigation argument or detailed material comparison. |

#### Quick recognition rules

| Question clue                                                    | Go first to                            | What Lecture 02 can do                                                                                      |
|------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------|
| “Climate vs Environment”                                         | Slide 2                                | Exact definitions and distinction.                                                                          |
| “Khana / south best / east second / north burden / west curse”   | Slides 9-11                            | Use the zoning sketch plus four climatic explanations.                                                      |
| “Khana in contemporary residence / functional zoning”            | Slides 9-12                            | Use traditional environmental logic as the evidence base; the final modern application must be synthesized. |
| “Vitruvius suggestions”                                          | Slides 6-7                             | Write the eight headings from Slide 7, preceded by the Vitruvius context on Slide 6.                        |
| “Designer responsibilities / complexities / essential knowledge” | Slides 13-14, then Lecture 04          | Lecture 02 frames the need; do not stop here. Lecture 04 is the direct source.                              |
| “Contemporary vs traditional materials / climate response”       | Slide 13, then later materials lecture | Only a broad “then vs now” comparison is present here; detailed material comparison is elsewhere.           |
| “Aristotle / Xenophon / ancient climate wisdom”                  | Slides 4-5                             | Good conceptual enrichment; no direct past question found in the supplied 2016-2024 bank.                   |
| “Vastu Shastra”                                                  | Slide 8                                | Visual reference only. No direct past-question match and little explanatory text.                           |

## 4. Slide-to-Question Reverse Index

Use this page while studying the lecture: after finishing a slide block, immediately practice the listed past questions. "Support" means the block helps the answer but is not sufficient by itself.

| Slides | Topic                                                             | Past questions                                        | Study note                                                                                                            |
|--------|-------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 2      | Climate and Environment definitions                               | 2017 Q5(a), Q6(a); 2018 Q1(a); 2024 Q5(a)             | High return: exact for Climate-vs-Environment; partial for questions that also require Context/Comfort.               |
| 3      | Is climate-responsive design a contemporary movement?             | No direct question found                              | Use only as a transition into ancient precedents; not standalone answer content.                                      |
| 4      | Aristotle: slope, east exposure, north-wind protection            | No direct question found                              | Conceptual support for climate-responsive planning; not directly tested in the supplied bank.                         |
| 5      | Xenophon: south aspect, winter sun, summer shade                  | No direct question found                              | Strong seasonal-solar example; use as enrichment, not as a substitute for the shading lecture.                        |
| 6-7    | Vitruvius and eight climatic suggestions                          | 2017 Q5(c)                                            | Exact, self-contained and easy to revise.                                                                             |
| 8      | Vastu Shastra                                                     | No direct question found                              | Graphic is present but explanatory text is minimal; do not invent details from the image alone.                       |
| 9-11   | KHANA maxim: zoning + environmental interpretation                | 2017 Q5(b); 2018 Q2(a); 2024 Q5(b)                    | Very high priority. These slides contain the actual layout, sketch and climatic reasons.                              |
| 12     | Application in contemporary residence                             | 2024 Q5(b)                                            | The slide only raises the application question. Use Slides 9-11 as evidence, then synthesize the contemporary layout. |
| 13     | Then vs Now: internal climate, cost, health, power/system failure | 2017 Q6(c); 2018 Q1(b); 2020 Q5(a), Q5(b); 2024 Q6(b) | Useful supporting framework for responsibility/mitigation/traditional-vs-modern questions; not complete by itself.    |
| 14     | "What should be the designer's task now?"                         | 2017 Q6(c); 2020 Q5(a); 2024 Q6(b)                    | Prompt only; the actual responsibilities and knowledge domains are in Lecture 04.                                     |
| 15     | References                                                        | No direct question                                    | Navigation only.                                                                                                      |

## 5. Coverage Gaps and Diagnostic Exclusions

The value of a mapping report is not only showing what a lecture covers, but also preventing you from forcing an incomplete lecture into a question it cannot answer.

| Gap                                                                                   | Affected questions                 | Action                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context and Comfort definitions/design framework                                      | 2017 Q5(a); 2018 Q1(a); 2024 Q5(a) | Lecture 02 only defines Climate/Environment. Use Lecture 01 for Climate-Context-Comfort.                                                                                                     |
| Formal designer responsibilities, essential knowledge and climate-design complexities | 2017 Q6(c); 2020 Q5(a); 2024 Q6(b) | Use Lecture 04. Slides 13-14 only explain why a designer must respond.                                                                                                                       |
| Exact "accept fate / reject development / mitigate" argument                          | 2018 Q1(b)                         | Use Lecture 01, which states the dilemma and "Design with Environment" response directly.                                                                                                    |
| Detailed contemporary vs traditional building-material comparison                     | 2020 Q5(b)                         | Slide 13 is a broad system-level comparison, not a material-property table. Use the later climate-friendly/material lecture.                                                                 |
| Contemporary application of KHANA beyond the traditional zoning logic                 | 2024 Q5(b)                         | Slides 9-11 give the vernacular environmental logic and Slide 12 asks for application, but no full modern plan is supplied. State clearly that the final layout is an application/synthesis. |
| Vastu Shastra explanation                                                             | No direct past question            | Slide 8 is mainly a diagram. Do not invent a detailed Vastu theory answer from this deck.                                                                                                    |
| Aristotle/Xenophon as standalone exam topics                                          | No direct 2016-2024 question found | Revise as conceptual support; prioritize Khana, Vitruvius and Climate-vs-Environment first.                                                                                                  |

|                    |                                                                                                                                                                                                                                                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mapping discipline | Not counted: the 2023 and 2024 heat-wave sustainability cases. Slide 13 has broad cost/health/power-failure themes, but the questions demand social, environmental and economic sustainability analysis that belongs to later lectures. This report counts only links where Lecture 02 provides usable answer content. |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



## 6. Recommended Study Plan

Study this lecture by exam return, not by slide order. The goal is to master the high-value self-contained blocks first, then use the remaining slides to strengthen reasoning and examples.

| Step | Study block                                                 | Slides    | Practice immediately                                                                           |
|------|-------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------|
| 1    | KHANA maxim: total plan + four directional climatic reasons | 9-11      | 2017 Q5(b); 2018 Q2(a)                                                                         |
| 2    | Contemporary KHANA application                              | 12 + 9-11 | 2024 Q5(b)                                                                                     |
| 3    | Climate vs Environment                                      | 2         | 2017 Q6(a); then use as partial support for 2017 Q5(a), 2018 Q1(a), 2024 Q5(a)                 |
| 4    | Vitruvius: context + eight suggestions                      | 6-7       | 2017 Q5(c)                                                                                     |
| 5    | Then vs Now + designer-task bridge                          | 13-14     | 2017 Q6(c); 2018 Q1(b); 2020 Q5(a), Q5(b); 2024 Q6(b) - then move to Lecture 01/04 as required |
| 6    | Aristotle + Xenophon ancient precedents                     | 4-5       | Use as evidence that climate-responsive design predates contemporary movements                 |
| 7    | Vastu Shastra visual reference                              | 8         | Low priority; no direct question found                                                         |

### Exam-ready minimum from Lecture 02

- Climate vs Environment: be able to reproduce the Slide-2 definitions and distinction.
- Vitruvius: memorize all eight suggestion headings from Slide 7.
- KHANA: redraw the four-direction zoning sketch and explain east pond, west bamboo, north low vegetation/banana and open south from the slide logic.
- Contemporary KHANA question: show that the traditional environmental insights are the evidence base; the modern plan is an application, not something already drawn in the deck.
- Designer responsibilities: recognize that Slides 13-14 are only the problem statement/bridge; move to Lecture 04 for the complete answer.
- Ancient experts: use Aristotle and Xenophon as supporting examples if asked whether climate-responsive design is a contemporary movement.
- Vastu: do not manufacture detailed theory from the diagram because the slide does not provide it.

**Final  
recommendation**

If time is short, prioritize Slides 9-12 first, then Slides 2 and 6-7. These blocks contain every Exact/Direct match in this lecture. Treat Slides 13-14 as a bridge to Lecture 04, not as a substitute for the full designer-responsibility answer.